Joseph Dwyer

Midlothian, VA | (717) 476 – 7196 | Joseph.Jerome.Dwyer@gmail.com | LinkedIn Profile

PhD: Computer Science | Research Focus: LLM Parameter Efficiency

KNOWLEDGE, SKILLS, & ABILITIES (KSA)

Primary Programming Languages:
C/C++ with STL and MPI, JAVA SE, SQL, Python

Operating Systems:
Linux

Integrated Development Environments:
VSCode, Jupyter Notebooks

Version Control:
Git/ GitHub

AI/ ML Tools:
Docker, Docker Compose,Vercel

Cloud Environments:
AWS/ SageMaker

PROFESSIONAL OVERVIEW

Computer Information Science (CIS) Associate Dean of Software Development | ECPI University | August 2025 – Present

Computer Information Science (CIS) Program Director | ECPI University | March 2016 – August 2025

Project Manager, Lead SharePoint Developer, Scrum Master | eTRANSERVICES, LLC | October 2015 – February 2016

Lead SharePoint Designer & Software Developer | Expert Consultants, Inc. | December 2012 – September 2014

EDUCATION & RESEARCH EXPERIENCE

Doctor of Philosophy: Computer Science | National University | 2025

Dissertation: A Quantitative Experimental Study on Exploring Parameter Efficiency with Varying Token Counts in OpenLLaMA Training

Investigates compute-efficient training strategies for LLMs. Grounded in scaling laws (Kaplan et al., Hoffmann et al.) and Big O Theory

Dissertation Available on ProQuest and Library of Congress

Master of Science in Computer Science | Hood College | Frederick, MD | 2010

Research: Parallel Programming, Fall 2008 | Presented an Implementation of the Gauss – Jordan Algorithm using MPI at CCSC Eastern Conference

Bachelor of Science in Mathematics | George Mason University | Fairfax, VA | 2005

Research: Long Range Interaction Potentials, Fall 2004 | Presented on the

Angular Dependence of the Asymptotic Long-Range Interaction Potentials Between CH_3OH-H_2

CERTIFICATIONS

Trainer the Trainer: Micro Credentials, Certified Scrum Master | Scrum Alliance | August 2026
Advanced Certified Scrum Master | Scrum Alliance | September 2022
|CompTIA IT Fundamentals (ITF+) | CompTIA **|** June 2020
MTA: Software Development, Fundamentals | Microsoft **|** May 2020
Security + CE | CompTIA | September 2015
Certificate of Professional Development: Client Programming | O'Reilly **|** January 2013
Sun Certified Associate, Java Platform, Standard Edition | Sun Microsystems **|** August 2010

DETAILED PROFESSIONAL EXPERIENCE

CIS Associate Dean of Software Development | East Coast Polytechnic Institute (ECPI) University | August 2026 – Present

- Implemented GitHub throughout the entire Software Development.
- Designed and implemented the Bachelor of Science in AI-Augmented Software Development degree.

CIS Program Director | East Coast Polytechnic Institute (ECPI) University
Glen Allen, Virginia **|** March 2016 – August 2026

- Facilitated and engaged with Henrico County Public Schools over the local Dual Enrollment program.
- Guided, mentored, and championed students through their Computer Information Systems degree.
- Garnered support within the local community over eSports presence; managed eSports arena.

Project Manager, Lead SharePoint Developer, Scrum Master | eTranservices, LLC
Fredericksburg, Virginia **|** October 2015 – February 2016

- Guided operations around Team Foundation Server; supported as a centralized code repository and project management tools and offered ad hoc reporting and presentation.
- Engaged in change management in support of clients; refined and automated the CCB Charter process.
- Optimized, initiated, and crafted standard operating procedures (SOP) over management, Software development and service desk ticket resolution processes; implemented Scrum methodologies, which improved communication between internal teams and external stakeholders.

Lead SharePoint Designer & Software Developer | Expert Consultants, Inc.
Frederick, Maryland **|** December 2012 – September 2014

- Mentored and trained junior designers; set design standards for revisions, provided team education, and acted as the main point of contact to disseminate information from management to implementation.
- Developed high-quality business solutions and delivered ahead of project deadlines; crafted multiple SharePoint solutions, which acted as the technical standard and exceeded expectations.
- Improved operations and efficiency with the development of a new tuition reimbursement feature under the company portal; utilized the following programs and systems for prototyping and database design: HSQL, SQL, Hibernate, Spring Frame Work, JSP, HTML, Junit, and more.

PUBLICATIONS & WORKS

Prior, Y., Dwyer, J., Martelli, M., & Bush, M. (2026, August). ***Understanding how full-timeworkers use AI: Patterns of use, trust, and perceived value*** **[Poster presentation].** American Psychological Association Annual Convention, Denver, CO, United States. ResearchGate.

Dwyer, J. (2026, June). ***A Quantitative Experimental Repeated Measures Study of Training Dynamics in a Small Llama Style Language Model Under a Compute-Aware Token Budget***. arXiv.

Dwyer, J. (2026, June). **Revisiting Training Scale: An Empirical Study of Token Count, Power Consumption, and Parameter Efficiency**. arXiv.

Dwyer, J. (2025, October). ***A Quantitative Experimental Study on Exploring Parameter Efficiency with Varying Token Counts in OpenLLaMA Training*****.** Dissertation, National University.

Prior, Y., West, J., & Dwyer, J. (2023, July). ***Technology and stress: A qualitative case study with Gen Z workers*****. Poster presented at the APA Annual Convention.** ResearchGate.

ACHIEVEMENTS

Leadership Institute Graduate | ECPI University | 2024

Faculty of the Year | ECPI University **|** 2023, 2017
